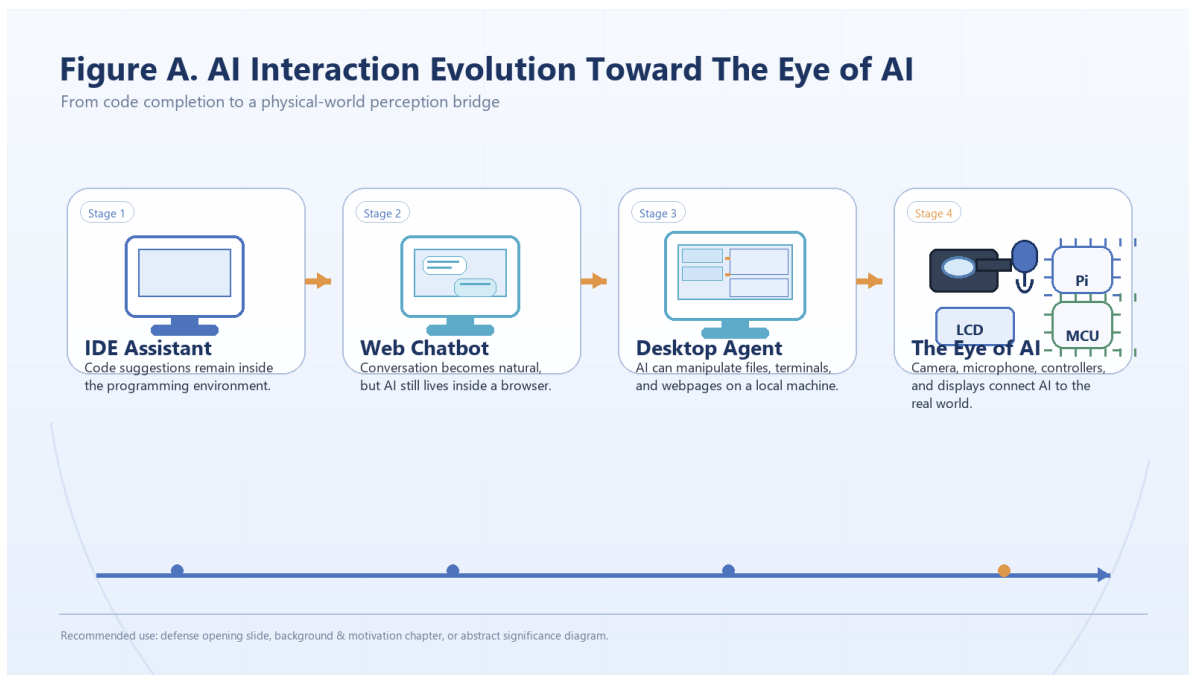


# 《The Eye of AI》答辩 PPT 逐页修改意见

对应文件：ai\_reality\_bridge\_english.pptx | 输出目标：逐页改到能撑住答辩主叙事与画面质量

生成时间：2026-07-02 | 供小组成员直接按页修改与补图使用



建议作为开篇总概念图使用：把“AI 从屏幕走向现实世界”的故事一次讲清楚。

## 一、这份 PPT 现在最大的结构性问题

### 总判断

- 现在的 PPT 技术内容并不差，但‘为什么这个项目值得讲’还没有压到最前面，老师第一眼看到的仍是一个标准课程项目，而不是一个有立意的 AI 原型。
- 整套 PPT 里多处出现重复英文标题、重复版式和重复画面风格，视觉类型不够丰富，容易让人觉得‘每页都差不多’。
- 中段有些页过于强调 course requirement mapping，而没有把这些课程点转译成‘为什么这样设计更

合理’。

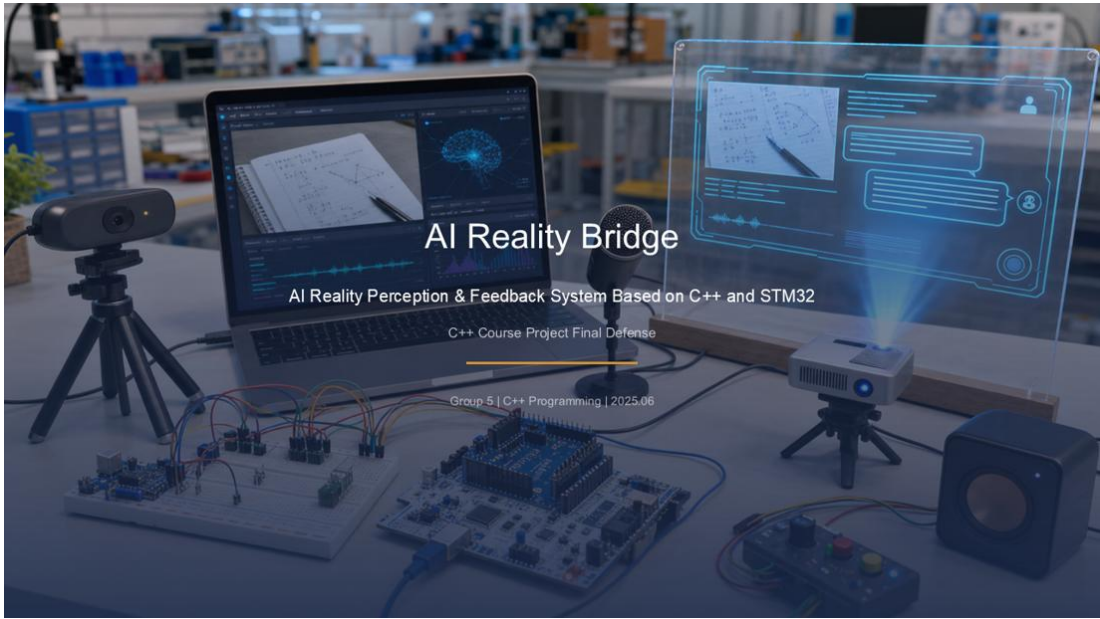
- 建议整体改成一条更像答辩故事线的结构：Why AI needs eyes → How our prototype builds the bridge → Why the C++/hardware architecture is credible → What has been validated and what remains future work.

### **建议整套 PPT 统一遵守的改法**

- 封面和背景部分必须把 AI 从 IDE 补全到桌面 Agent 的演进讲出来，再落到 ‘AI 仍被困在电脑里，所以我们做 The Eye of AI’。
- 所有出现 smart glasses 的地方，都要加一句 prototype / prototype validation，避免被误听成已经做出成熟眼镜产品。
- 若台上演示用最新版系统，请把 AI 架构统一成：Qwen for perception, DeepSeek for final reasoning and answer composition。
- 每一节都要尽量做到 ‘一页一主图 + 三到五个点’，避免一页里塞满表格与小字。

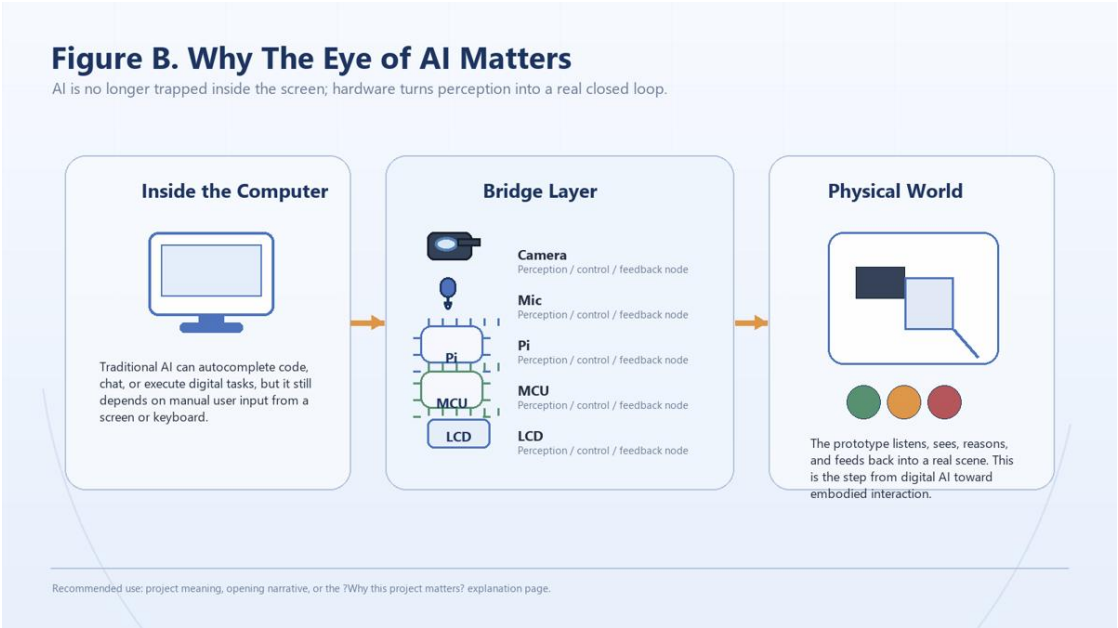
二、逐页修改意见

第 1 页：封面页



当前第 1 页截图

| 栏目       | 建议                                                                                                             |
|----------|----------------------------------------------------------------------------------------------------------------|
| 建议标题/主句  | The Eye of AI   A Prototype Bridge Connecting Agentic AI to Physical Reality                                   |
| 这一页现在的问题 | 标题还是 AI Reality Bridge，不够像一个有概念与愿景的项目。；副标题过于课程化，缺少 ‘桥梁意义’ 和 ‘prototype’ 这两个关键词。                                |
| 建议怎么改    | 把主标题改成 The Eye of AI。；副标题改成 A Prototype Bridge Connecting Agentic AI to Physical Reality。；页脚保留课程信息即可，不要和主标题抢戏。 |
| 答辩时怎么讲   | 开场第一句话就说：AI 已经学会写代码、聊天、操作电脑，但它还没有真正看见现实世界。；接着引出：The Eye of AI 就是为它搭一座桥。                                        |



建议作为封面主视觉或封面后第二页主图。

第 2 页：目录页

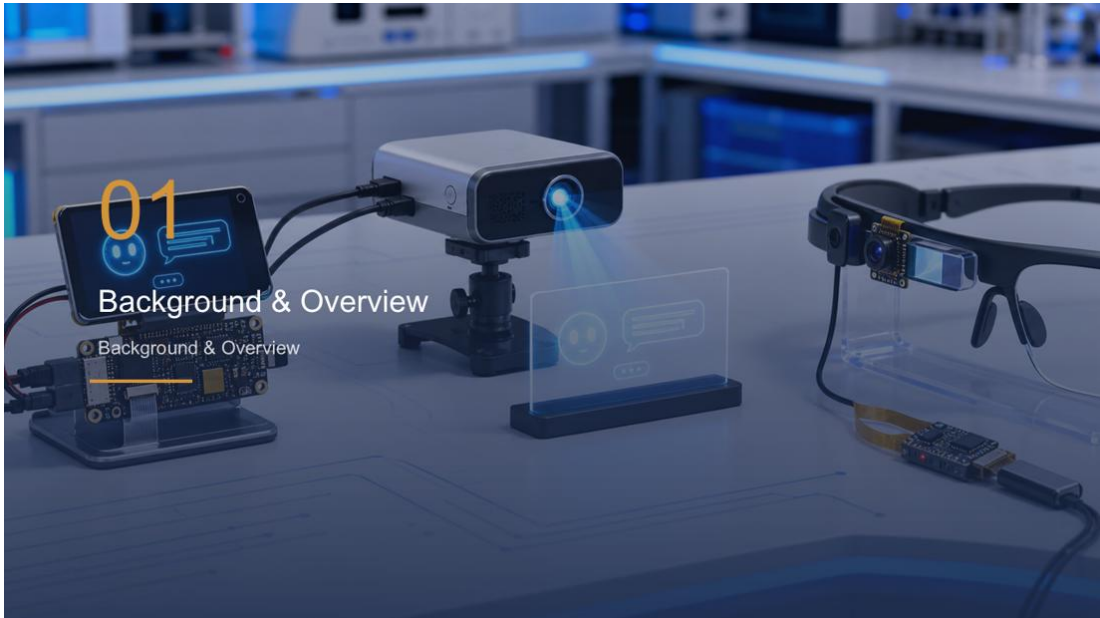
|                             |                                                                  |
|-----------------------------|------------------------------------------------------------------|
| <b>CONTENTS</b><br>CONTENTS | <b>01</b> Background & Overview<br>Background & Overview         |
|                             | <b>02</b> Architecture & C++ Design<br>Architecture & C++ Design |
|                             | <b>03</b> Testing & Validation<br>Testing & Validation           |
|                             | <b>04</b> Conclusion & Team<br>Conclusion & Team                 |

当前第 2 页截图

|    |    |
|----|----|
| 栏目 | 建议 |
|----|----|

|          |                                                                                                                                      |
|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| 建议标题/主句  | CONTENTS   From Screen-Bound AI to The Eye of AI                                                                                     |
| 这一页现在的问题 | 双语标题重复，层次感不强。；目录只是在列章节，没有把答辩故事线讲出来。                                                                                                  |
| 建议怎么改    | 把 4 个章节名字改成更有叙事感的版本，如 Why AI Needs Eyes / Prototype Bridge Design / C++ & Hardware Realization / Validation & Outlook。；保留编号，但减少重复排版。 |
| 答辩时怎么讲   | 告诉老师：我们先讲项目为什么成立，再讲系统如何落地，最后讲验证与边界。                                                                                                  |

第 3 页：章节分隔页 01



当前第 3 页截图

| 栏目       | 建议                     |
|----------|------------------------|
| 建议标题/主句  | 01   Why AI Needs Eyes |
| 这一页现在的问题 | 分隔页功能可以，但目前太空、太重复。     |

|        |                                                                              |
|--------|------------------------------------------------------------------------------|
| 建议怎么改  | 保留分隔页，但加一句 section thesis，比如：Why current AI still lacks a bridge to reality. |
| 答辩时怎么讲 | 分隔页要承担 ‘换气’ 作用，不只是页码。                                                        |

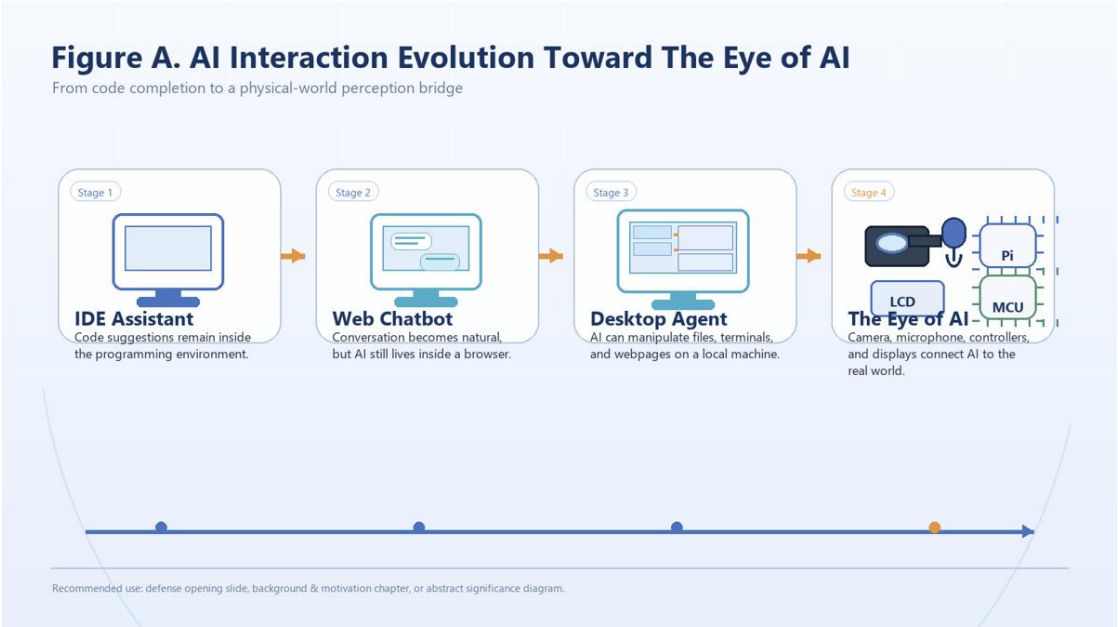
第 4 页：Background & Motivation



当前第 4 页截图

| 栏目       | 建议                                                                                                                                             |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 建议标题/主句  | Project Background & Motivation   From Screen-Bound AI to The Eye of AI                                                                        |
| 这一页现在的问题 | 重点放在传统交互缺陷，缺少最能拔高项目的 AI 演化逻辑。；这一页文字太多，老师不容易一眼抓到主线。                                                                                             |
| 建议怎么改    | 改成上半页 ‘AI evolution timeline’，下半页 ‘screen-bound limitation + our bridge response’。；四个传统缺陷可以缩成右侧 4 个短标签，不要占据整页。；把 project motivation 单独放成一条强论断。 |

|        |                                                                                    |
|--------|------------------------------------------------------------------------------------|
| 答辩时怎么讲 | <p>先讲：AI 会写代码、会聊天、会操作文件，但不会自然地感知你眼前的世界。；再讲：所以我们不是做 another chatbot，而是做 AI 的眼睛。</p> |
|--------|------------------------------------------------------------------------------------|

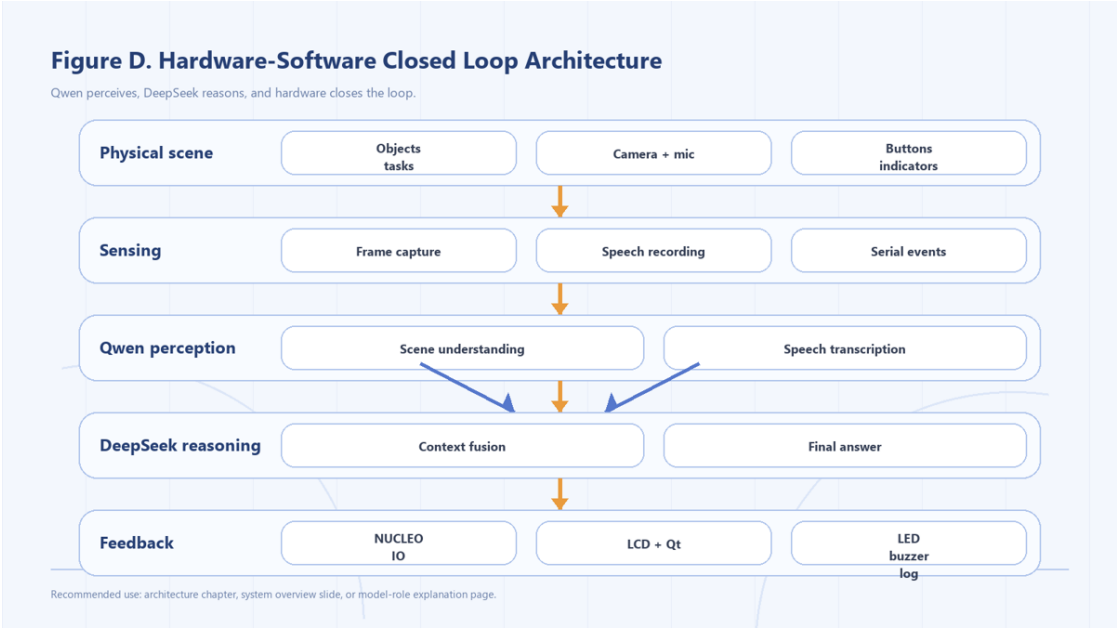


这页最应该用时间线图替代现在的大段文字。

## 第 5 页：System Goals & Objectives



| 栏目       | 建议                                                                                                                                                                                 |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 建议标题/主句  | System Goals   Validate a Complete Perception–Reasoning–Feedback Loop                                                                                                              |
| 这一页现在的问题 | 功能目标和课程考核点混写在一起，显得不够高级。； AI 模型角色尚未跟当前项目版本完全对齐。                                                                                                                                     |
| 建议怎么改    | 把内容拆成 Capability Goals / Prototype Goals / Course Implementation Goals 三组。； 把 Qwen 写成 perception layer，把 DeepSeek 写成 final reasoning layer（若答辩演示使用当前版本）。； 课程点只保留在每行最右侧的小标签里，弱化存在感。 |
| 答辩时怎么讲   | 这一页不是功能罗列，而是说明 ‘我们到底想证明什么’ 。                                                                                                                                                       |

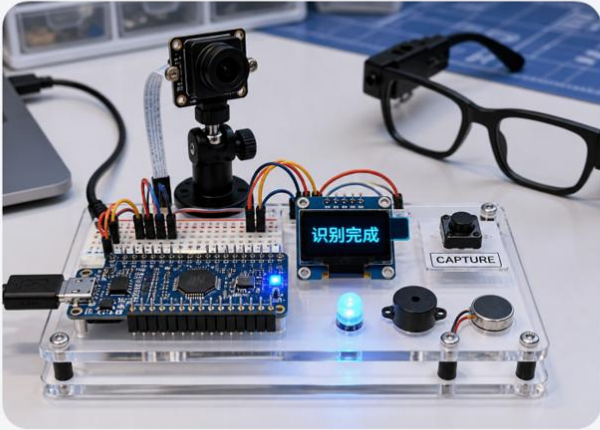


这张图可以和能力目标列表并排放。

第 6 页 : Hardware Composition

01 Background & Overview

Hardware Composition / Hardware Composition



- Raspberry Pi 5 — Linux Host, Camera, Mic, Network, AI API
- NUCLEO-F446RE — MCU Controller, GPIO, UART, I2C
- C270 USB Camera — 1280×720 Image Capture
- OLED Display — Local Status Output (128×64)
- Three-Key Keyboard — Trigger Buttons
- LED + Buzzer + Vibration — Multi-Modal Feedback
- Windows PC — Qt 6 Control Center

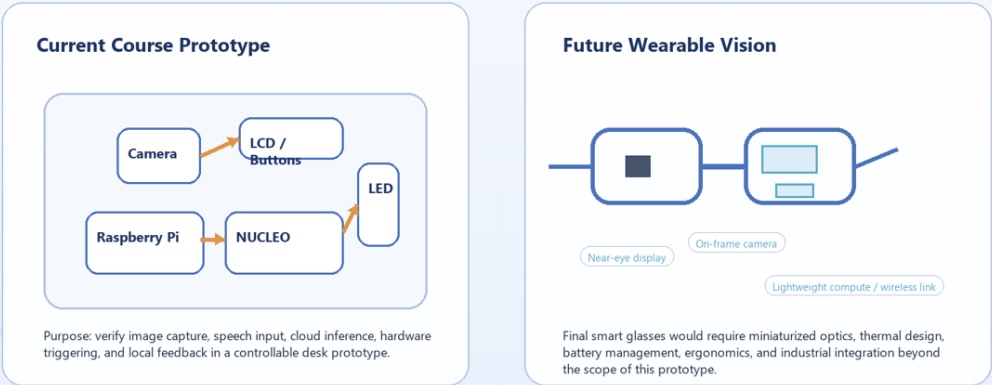
06

当前第 6 页截图

| 栏目       | 建议                                                                                      |
|----------|-----------------------------------------------------------------------------------------|
| 建议标题/主句  | Prototype Hardware Composition   Functional Validation over Product Form                |
| 这一页现在的问题 | 像硬件清单页，没有把 prototype 的角色讲透。；老师容易误解为我们真的在做成品眼镜。                                          |
| 建议怎么改    | 每个硬件后面不是写规格，而是写它对应的功能角色。；页底必须加一句 prototype clarification。；加一张 prototype vs product 对照图。 |
| 答辩时怎么讲   | 强调：我们追求的是功能语义闭环，不是工业外观定型。                                                               |

Figure C. Prototype Validation vs. Final Smart-Glasses Product

The current project validates capability, not final industrial form.



Recommended use: clarify that the project is a prototype implementing smart-glasses functions, not a finished smart-glasses product.

非常适合作为这页的第二视觉锚点。

第 7 页：章节分隔页 02

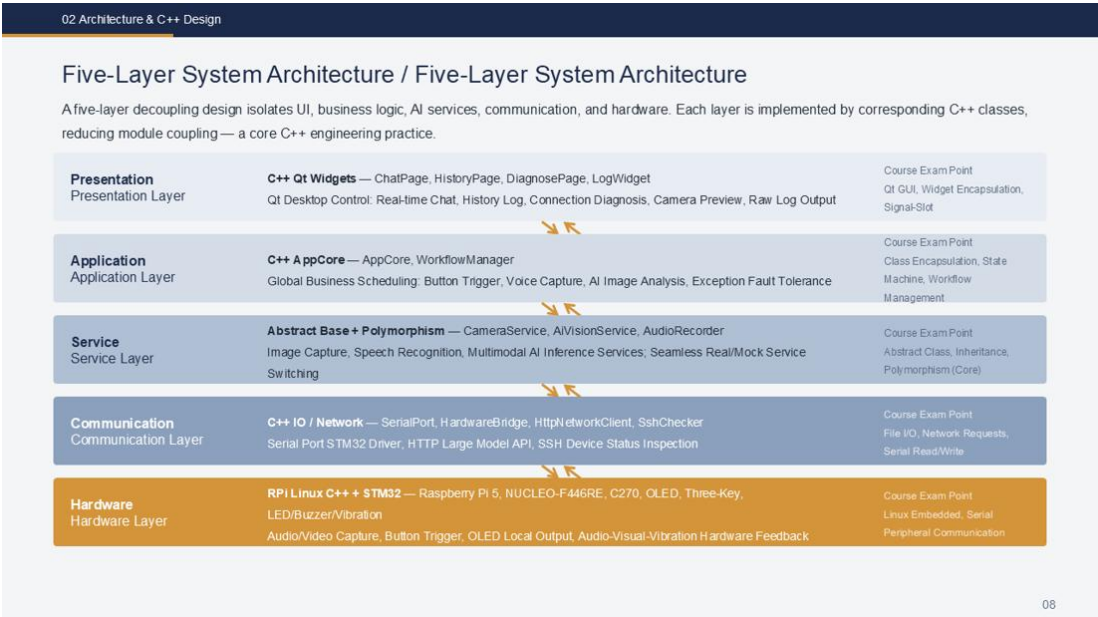


当前第 7 页截图

| 栏目 | 建议 |
|----|----|
|----|----|

|          |                                                                        |
|----------|------------------------------------------------------------------------|
| 建议标题/主句  | 02   How the Bridge Is Built                                           |
| 这一页现在的问题 | 与前一张分隔页风格过于重复。                                                         |
| 建议怎么改    | 加一句本章主题：How C++ architecture turns the idea into a reliable prototype. |
| 答辩时怎么讲   | 让老师知道下面会进入 ‘如何实现’ 而不是继续讲意义。                                            |

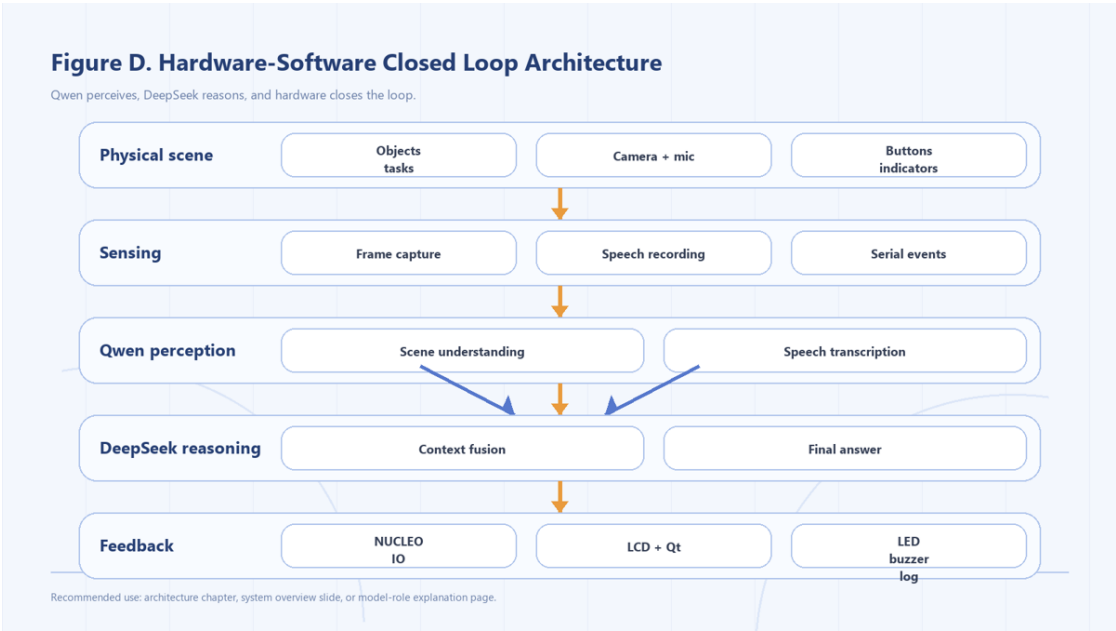
第 8 页：Five-Layer System Architecture



当前第 8 页截图

| 栏目       | 建议                                                                               |
|----------|----------------------------------------------------------------------------------|
| 建议标题/主句  | Five-Layer Architecture   Decoupling Perception, Reasoning, I/O, and Supervision |
| 这一页现在的问题 | 信息很多，但考试点标签过多，削弱了工程叙事。；五层架构的价值没有被一句话说透。                                          |

|        |                                                                    |
|--------|--------------------------------------------------------------------|
| 建议怎么改  | 保留五层，但把每层缩成一句角色描述。；把 exam point 字样做成小角标，别成为视觉主角。；用一张更强的架构主图替换大段说明。 |
| 答辩时怎么讲 | 强调这套分层是为了解耦：感知、推理、通信、硬件和显示可以独立演化。                                  |



这张图最适合做这页的主图。

第 9 页 : Layer Code Mapping

02 Architecture & C++ Design

Layer Code Mapping / Layer Code Mapping

The five-layer architecture maps to independent code directories. Each layer encapsulates dedicated C++ classes with header/source decoupling, strictly following OOP encapsulation, inheritance, and polymorphism principles.

| Layer                                | Directory             | Core C++ Classes                                                                                                      | Key Coding Capabilities                                                                    |
|--------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Presentation<br>Presentation Layer   | ui/                   | ChatPage, HistoryPage, DiagnoseWidget, LogViewer                                                                      | Qt6 Custom Widgets, Signal-Slot, Window Paging, Real-time Image Rendering                  |
| Application<br>Application Layer     | app/                  | AppCore, WorkflowManager, FaultHandler                                                                                | Business State Machine, Fault Tolerance, Multi-module Coordination, Lifecycle Management   |
| Service<br>Service Layer             | service/              | CameraService, AVisionService, AudioRecorder (abstract)<br>OpenCvCameraService, QwenVisionService, MockAVisionService | Abstract Interface, Inheritance Override, Runtime Polymorphism, Module Hot-Swap            |
| Communication<br>Communication Layer | comm/                 | SerialPort, HardwareBridge, HttpClient                                                                                | Binary Serial Read/Write, libcurl Network Requests, Data Pack/Unpack, IO Exception Capture |
| Hardware<br>Hardware Layer           | rpi_capture/ + stm32/ | RpiCapture, Stm32Driver                                                                                               | Linux Low-level Device Access, Cross-device Serial Communication, Peripheral GPIO Control  |

Engineering Code Standards

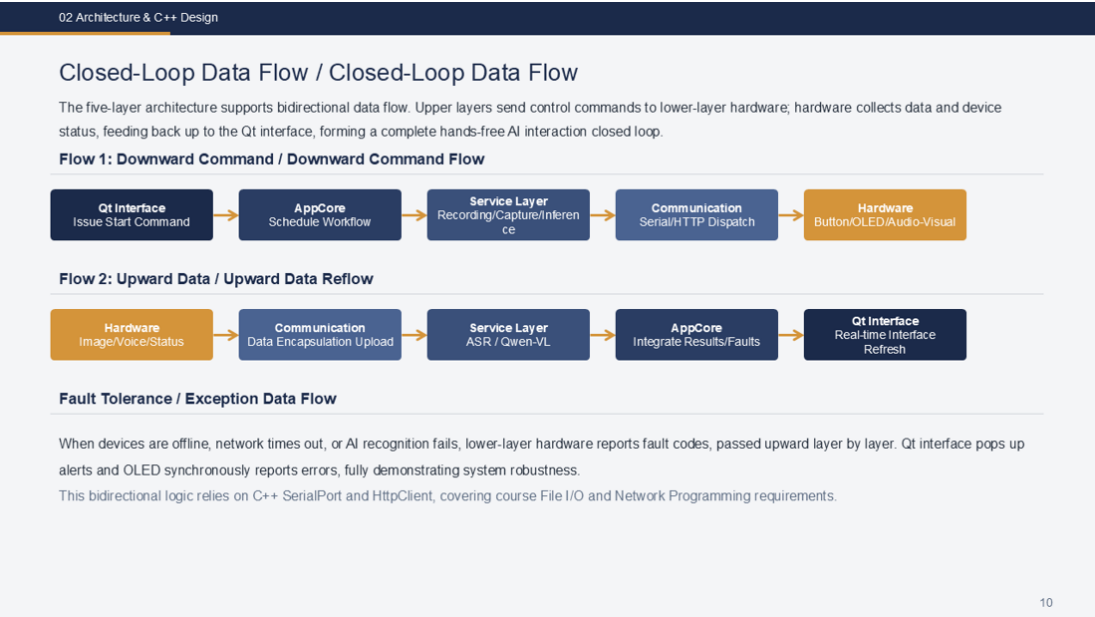
Header (.h) declares classes/interfaces, Source (.cpp) implements logic, compilation decoupled. • All services depend on abstract base classes, not concrete implementations, conforming to the Open-Closed Principle.  
• Unified exception capture mechanism, custom exception classes per layer, hierarchical fault reporting. • AuditLogStore independently encapsulated, globally called, fulfilling the course random file I/O requirement.

09

当前第 9 页截图

| 栏目       | 建议                                                                                                                       |
|----------|--------------------------------------------------------------------------------------------------------------------------|
| 建议标题/主句  | Code Mapping   How the Architecture Becomes C++ Engineering                                                              |
| 这一页现在的问题 | 这一页文字密度高，阅读负担大。；目录映射有价值，但不需要全部铺开。                                                                                        |
| 建议怎么改    | 保留最能证明 C++ 工程性的 4 组类/目录映射即可。；把其余细节挪到附录或答辩口述。；突出 header/source separation, abstract interfaces, AuditLogStore random I/O。 |
| 答辩时怎么讲   | 这页要服务课程要求，但不要让它拖慢答辩节奏。                                                                                                   |

第 10 页 : Closed-Loop Data Flow

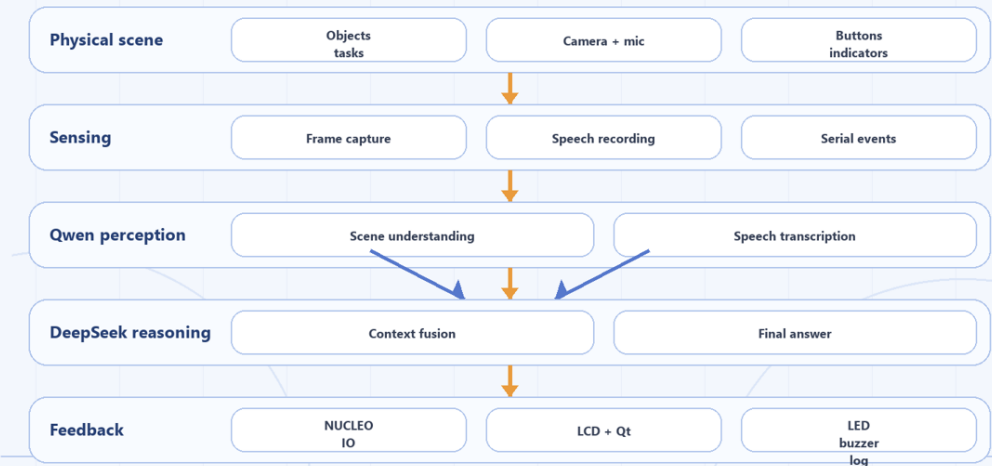


当前第 10 页截图

| 栏目       | 建议                                                                                                                   |
|----------|----------------------------------------------------------------------------------------------------------------------|
| 这一页现在的问题 | 上下行数据流分开写是对的，但现在还是过于文字化。；没有一眼看出‘闭环’。                                                                                 |
| 建议怎么改    | 改成一个环形或双向箭头闭环图：hardware trigger -> perception -> reasoning -> feedback -> supervision。；异常流只保留 1 条 fault propagation。 |
| 答辩时怎么讲   | 这里要让老师一眼看到：这是一个闭环系统，不是单向调用。                                                                                          |

Figure D. Hardware-Software Closed Loop Architecture

Qwen perceives, DeepSeek reasons, and hardware closes the loop.



Recommended use: architecture chapter, system overview slide, or model-role explanation page.

可把这张图局部改成更强调双向流。

第 11 页 : Complete Interaction Flow

02 Architecture & C++ Design

### Complete Interaction Flow / Complete Interaction Flow

The entire interaction is triggered by a single physical button press, no touchscreen needed. A three-level fault fallback strategy ensures stable operation in noisy, disconnected, or peripheral-offline scenarios, solving the pain points of cumbersome traditional AI interaction with poor fault tolerance.

Three-Key Trigger  
Physical Button Trigger

5s Voice Recording  
Audio Recording

ASR Speech-to-Text  
Speech-to-Text

Camera Snapshot  
Real-time Image Capture

Qwen-VL Analysis  
Multimodal AI Understanding

OLED + Qt Sync  
Synchronize Hardware and Interface

#### Fallback Fault Tolerance Strategy / Fault Tolerance Fallback Mechanism

**Speech Recognition Failure**  
Switch to manual text input channel to avoid no-input freeze and ensure interaction chain continuity.

**Camera Offline**  
Load locally cached historical images to ensure AI analysis chain continuity and support continuous use.

**Network Connection Interrupted**  
Hardware audio-visual alarm + OLED fault code printing for quick fault-level localization and troubleshooting.

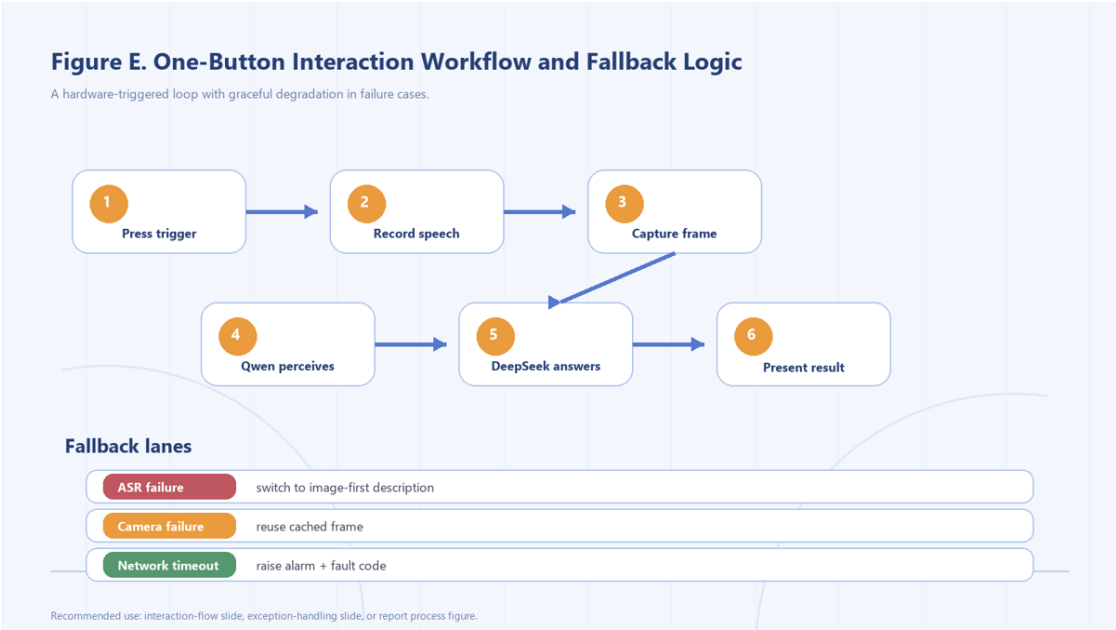
11

当前第 11 页截图

| 栏目 | 建议 |
|----|----|
|----|----|

The Eye of AI | 答辩 PPT 修改意见

|          |                                                                                   |
|----------|-----------------------------------------------------------------------------------|
| 建议标题/主句  | One-Trigger Interaction Flow   Main Path and Fallback Path                        |
| 这一页现在的问题 | 主流程与 fallback 都有，但字还是太多。；缺少 ‘单次演示怎么走完’ 的直观感。                                      |
| 建议怎么改    | 把主流程画成 storyboard：按键 -> 录音 -> 识别 -> 拍照 -> 感知 -> 回答 -> LCD/Qt。；fallback 做成右侧三条短策略。 |
| 答辩时怎么讲   | 这是现场演示页，老师看到这里应当知道 ‘按一下会发生什么’ 。                                                   |



这页最适合直接用主流程 + fallback 示意图。

第 12 页 : C++ Object-Oriented Design

02 Architecture & C++ Design

C++ Object-Oriented Design / C++ Object-Oriented Design

This project strictly follows C++ OOP standards. **Pure virtual abstract interface classes** define unified service standards, with different business implementations completed through inheritance and polymorphism, fully covering the core OOP assessment content.

**Abstract Interface / Abstract Base Class**

CameraService  
Pure Virtual Interface capture()

AIVisionService  
Pure Virtual Interface analyze()

AudioRecorder  
Pure Virtual Interface record()

**Derived Implementation / Derived Implementation Classes**

OpenCvCameraService  
OpenCV Real Camera Capture

QwenVisionService  
Call Qwen API

MockAIVisionService  
Offline Simulated AI Return

ShellAudioRecorder  
System Interface Audio Recording

**Design Benefits / Design Benefits**

① **Satisfies Course Polymorphism Assessment**

Relies on C++ abstract classes, inheritance, and dynamic binding to achieve polymorphism — the core project practice.

② **Decoupling and Parallel Development**

Upper layers only depend on abstract interfaces, not binding to hardware/API. Teams can develop in parallel without interference.

③ **Supports Offline Unit Testing**

MockAIVision requires no network or keys, enabling full business process debugging without hardware.

④ **Modules Can Be Quickly Replaced**

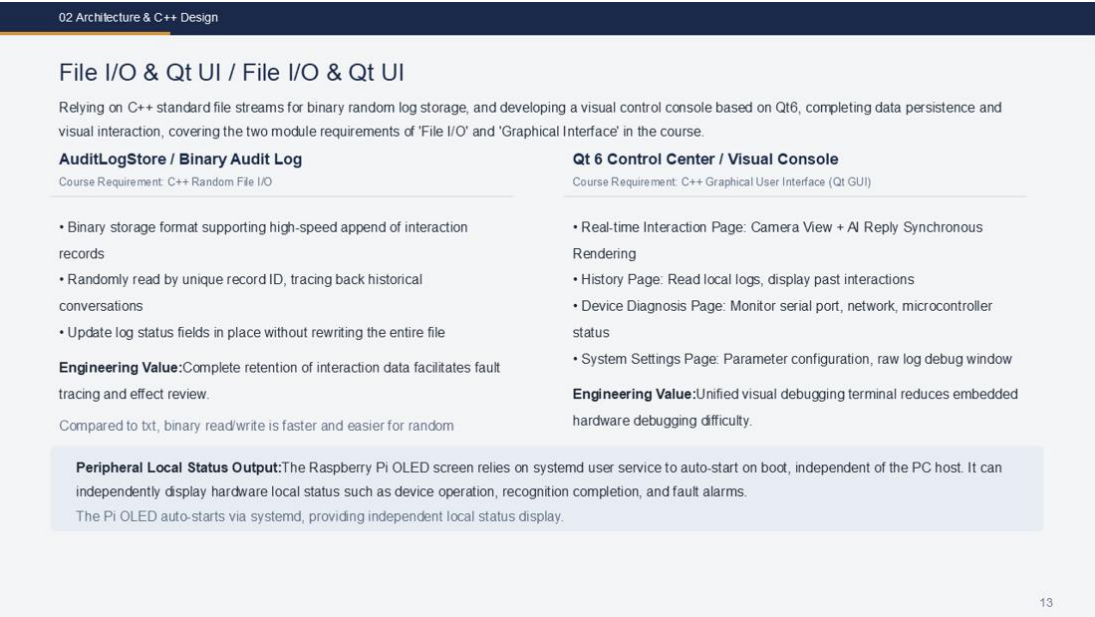
Only new derived classes need to be added when replacing cameras or models, without modifying upper-layer code.

当前第 12 页截图

| 栏目       | 建议                                                                                               |
|----------|--------------------------------------------------------------------------------------------------|
| 建议标题/主句  | C++ OOP Design   Abstract Interfaces for Real and Mock Services                                  |
| 这一页现在的问题 | 课程要求满足得不错，但还是偏 ‘证明我学了 OOP’ ；没有把测试性、可替换性这些工程收益讲深。                                                 |
| 建议怎么改    | 把 benefits 排成 ‘testability / replaceability / decoupling / offline debug’ 四点。；类图只保留最关键的抽象接口和派生类。 |
| 答辩时怎么讲   | 强调：多态在这里不是形式主义，而是稳定演示和替换服务的前提。                                                                   |

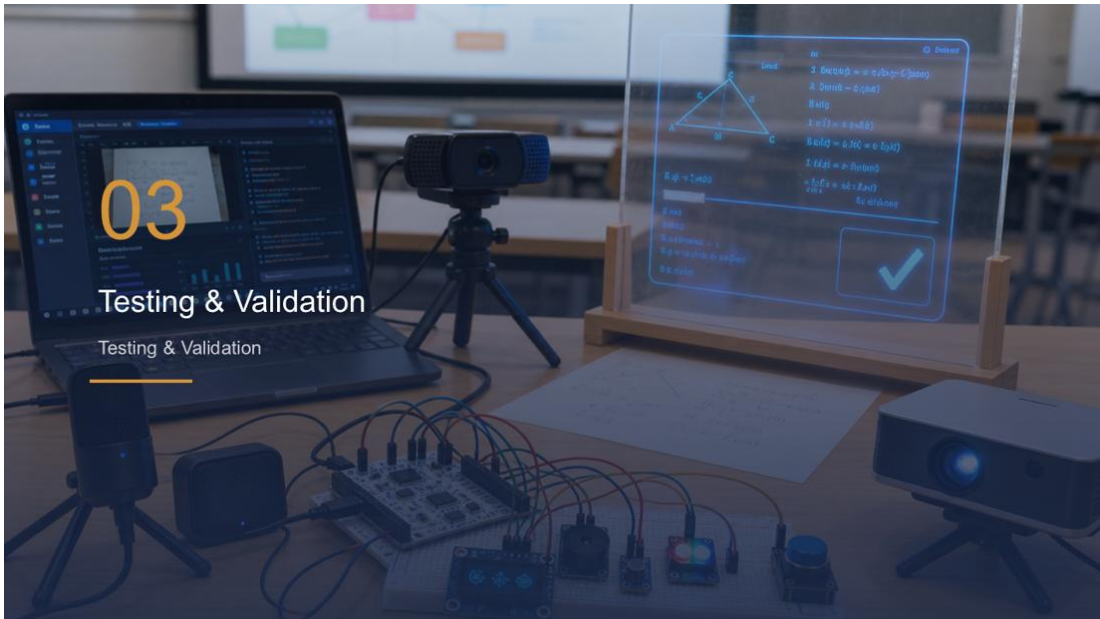
The Eye of AI | 答辩 PPT 修改意见

第 13 页 : File I/O & Qt UI



当前第 13 页截图

| 栏目       | 建议                                                                                                                                            |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 建议标题/主句  | Random File I/O and Qt Workbench   Persistence Meets Supervision                                                                              |
| 这一页现在的问题 | 这一页把 file I/O、Qt UI、OLED auto-start 都挤在一起，容易散。；老师会觉得这页像 ‘三页内容拼一页’ 。                                                                           |
| 建议怎么改    | 拆成左侧 AuditLogStore random access，右侧 Qt control center / OLED output. ；把 binary random I/O 的工程意义说清楚：history replay、fault tracing、auditability。 |
| 答辩时怎么讲   | 这页要说明：课程要求并没有脱离系统目标，而是被吸收到 Agent 体验里。                                                                                                         |



当前第 14 页截图

| 栏目       | 建议                                                                                      |
|----------|-----------------------------------------------------------------------------------------|
| 建议标题/主句  | 03   Validation and Boundaries                                                          |
| 这一页现在的问题 | 和前面分隔页一样，存在视觉重复。                                                                        |
| 建议怎么改    | 加一句 section thesis: What exactly has been validated, and where does the prototype stop? |
| 答辩时怎么讲   | 把测试与边界感讲出来，而不是只讲‘我们跑通了’。                                                                |

第 15 页 : Testing Matrix & Exception Handling

03 Testing & Validation

Testing Matrix & Exception Handling / Testing & Exception Handling

Functional Testing Matrix / Functional Testing Matrix

| Test Item       | Input        | Expected            | Result | C++ Class                   | Testing Purpose                                                                                        |
|-----------------|--------------|---------------------|--------|-----------------------------|--------------------------------------------------------------------------------------------------------|
| Serial Port     | /dev/ttyACM0 | Bi-directional data | ✓ Pass | SerialPort / HardwareBridge | Verify serial port encapsulation class, implementing command interaction between MCU and host computer |
| Camera          | C270 USB     | 1280×720 JPEG       | ✓ Pass | OpenCvCameraService         | Verify OpenCV image capture encapsulation, meeting real-time capture requirements                      |
| Voice Recording | 5s trigger   | WAV file            | ✓ Pass | ShellAudioRecorder          | Verify audio I/O encapsulation, implementing hands-free voice input                                    |
| ASR             | WAV upload   | Chinese text        | ✓ Pass | QwenAsrService              | Verify speech recognition, replacing manual text input                                                 |
| Qwen-VL         | Image + text | JSON result         | ✓ Pass | QwenVisionService           | Verify large model API calls, implementing image-text joint analysis                                   |
|                 |              |                     |        |                             |                                                                                                        |

Exception Handling / Exception Handling

- Serial port device change → Auto-scan /dev/ttyACM\* and reconnect
- Camera unreadable → Capture exception, use the last cached frame
- ASR service failure → Degrade to text prompt input without interrupting the main process
- Network disconnection → Red LED flashing + OLED error code + Qt status bar prompt
- API timeout → Exponential backoff retry, up to 3 times
- Chinese encoding → Force UTF-8 end-to-end processing

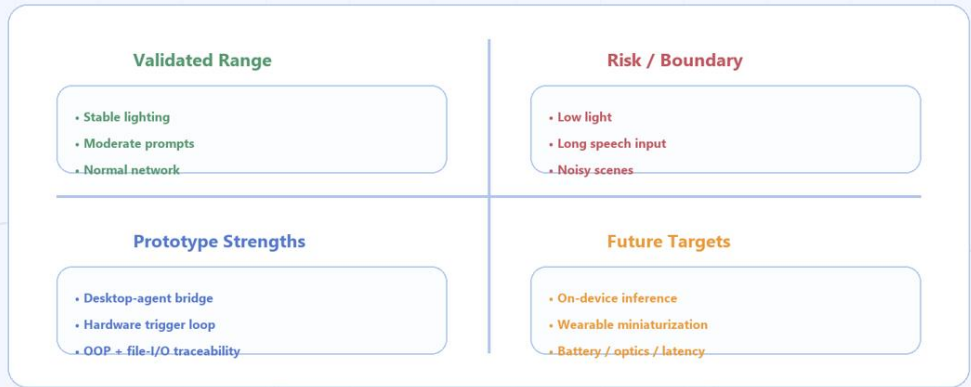
16

当前第 15 页截图

| 栏目       | 建议                                                                                |
|----------|-----------------------------------------------------------------------------------|
| 建议标题/主句  | Testing & Boundary Validation   What the Prototype Can Reliably Do                |
| 这一页现在的问题 | 异常处理很好，但仍主要是文字表。；缺少 ‘边界图’ 和 ‘我们究竟证明了什么’ 。                                         |
| 建议怎么改    | 在左侧给出 function/system testing，右侧给出 boundary & fault tolerance。；把异常流和边界流合并成更强的证据链。 |
| 答辩时怎么讲   | 老师在这里最关心的不是 pass，而是系统有没有边界意识。                                                     |

Figure F. Prototype Boundary Conditions and Validation Scope

Use this figure to state clearly what is validated, what is risky, and what remains future work.



Recommended use: testing chapter, limitation statement, or defense Q&A preparation page.

建议直接把边界条件视觉化，替代纯文字异常列表。

第 16 页：章节分隔页 04



当前第 16 页截图

| 栏目 | 建议 |
|----|----|
|----|----|

|          |                                                                    |
|----------|--------------------------------------------------------------------|
| 建议标题/主句  | 04   Meaning, Validation, and Next Step                            |
| 这一页现在的问题 | 结尾章节切换页偏空。                                                         |
| 建议怎么改    | 加一句：What does this project finally mean, and where can it go next? |
| 答辩时怎么讲   | 为结尾升华预热。                                                           |

第 17 页：Course Requirements & Team Mapping

04 Conclusion & Team

Course Requirements & Team Mapping / Course Requirements & Team Mapping

Course Requirements Mapping / Course Requirements Mapping

| Requirement                | Implementation                                                                             | Evidence                 |
|----------------------------|--------------------------------------------------------------------------------------------|--------------------------|
| UI Interface               | Qt 6 Windows Control Center + Raspberry Pi LCD local display                               | Screenshots + LCD photos |
| Classes & Objects          | App, HardwareBridge, SerialPort, QwenVisionService, etc.                                   | Class diagram + code     |
| Inheritance & Polymorphism | CameraService/OpenCvCameraService, AI/VisionService/QwenVisionService/MockAI/VisionService | Inheritance graph        |
| Random File I/O            | AuditLogStore binary log: append, read, update-by-ID                                       | Log file + history page  |
| Code Scale                 | Bridge + NUCLEO firmware + Qt center, >2000 lines                                          | Code statistics          |

Team Task Completion / Team Task Completion Table

| Member   | Backend     | Qt UI       | Embedded    | AI API      | Document    |
|----------|-------------|-------------|-------------|-------------|-------------|
| Member A | ✓ Completed | ✓ Completed | —           | —           | ✓ Completed |
| Member B | —           | —           | ✓ Completed | ✓ Completed | —           |
| Member C | ✓ Completed | ✓ Completed | ✓ Completed | —           | —           |
| Member D | —           | —           | ✓ Completed | ✓ Completed | ✓ Completed |

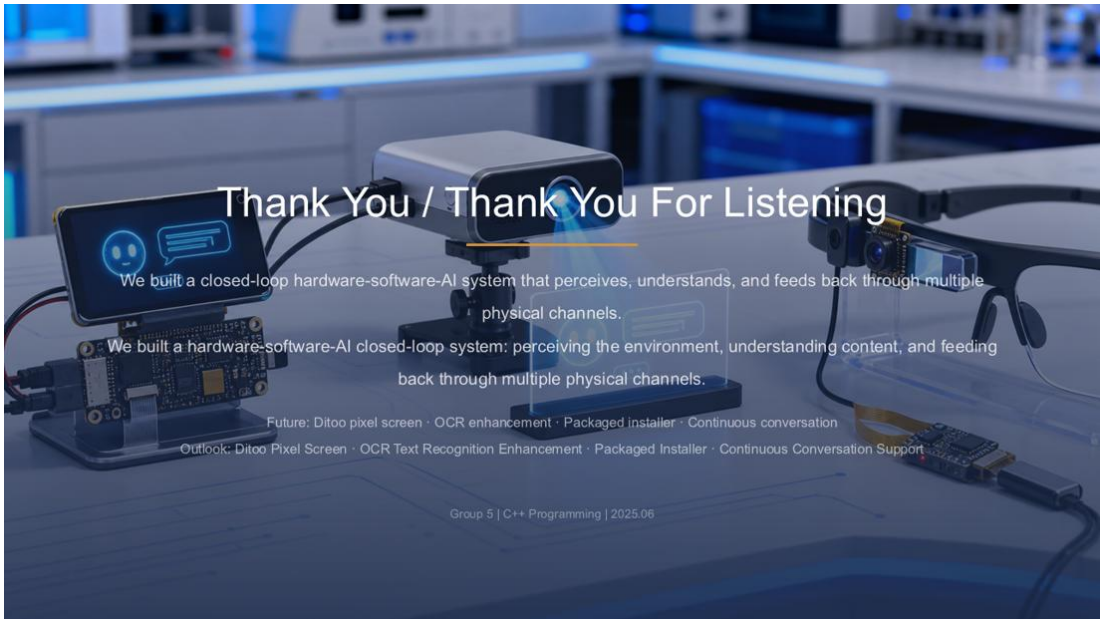
15

当前第 17 页截图

| 栏目       | 建议                                                                                           |
|----------|----------------------------------------------------------------------------------------------|
| 建议标题/主句  | Course Requirement Coverage   Why This Is a Real C++ Systems Project                         |
| 这一页现在的问题 | 这一页对课程要求交代充分，但放在主线里会显得比较“答作业”。；团队映射与课程映射挤在一起，视觉负担大。                                          |
| 建议怎么改    | 主答辩里只保留“requirements satisfied”精简版。；团队分工表建议移动到附录或口头一带而过。；把这一页改成“why this counts as a serious |

|        |                                 |
|--------|---------------------------------|
|        | C++ course project’ 。           |
| 答辩时怎么讲 | 这页只需要让老师安心：课程指标全部真实落地，而不是主要展示页。 |

第 18 页：Thank You / Future

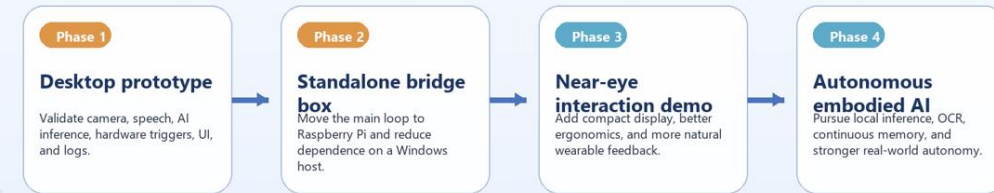


当前第 18 页截图

| 栏目       | 建议                                                                                                                                                                  |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 建议标题/主句  | Conclusion & Outlook   From Prototype Bridge to Future Embodied AI                                                                                                  |
| 这一页现在的问题 | 结尾 still works，但没有把 ‘AI 之眼’ 的意义钉住。；未来方向有，但缺少结构性路线图。                                                                                                                 |
| 建议怎么改    | 在 Thanks 前加一句 concluding thesis: We did not build finished smart glasses; we proved that AI can be given a practical bridge to reality.; 未来方向改成 roadmap，不要只是一行愿望列表。 |
| 答辩时怎么讲   | 最后一句建议收在：The Eye of AI is the moment when a desktop agent starts to see the world outside the screen.                                                               |

## Figure G. Research Roadmap After the Current Prototype

From a desk prototype to a stronger embodied AI system.



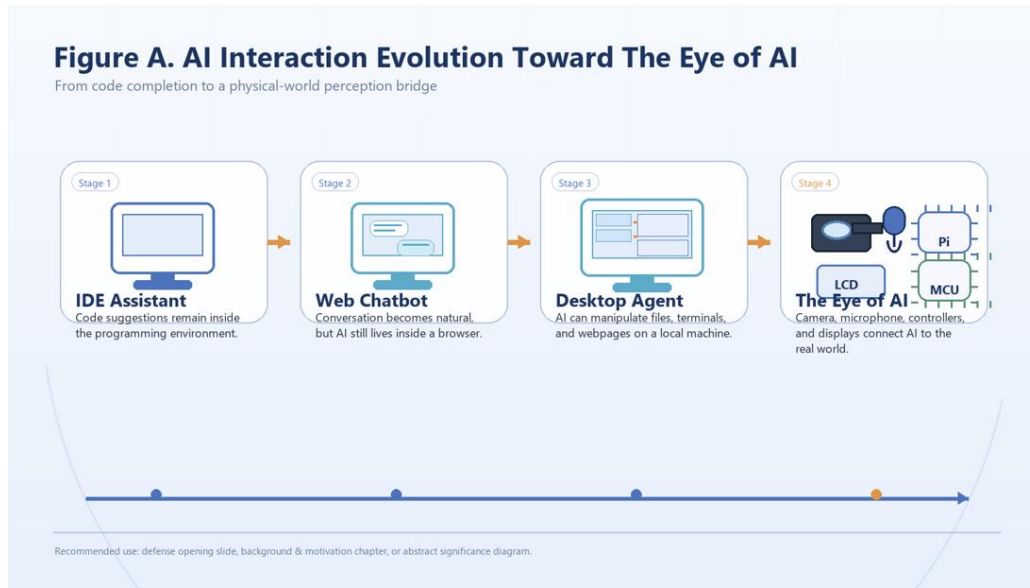
Recommended use: conclusion slide, future work section, or the final summary of the written report.

结尾页最适合放 roadmap，增强完整度。

### 三、建议直接补进 PPT 的图像资源索引

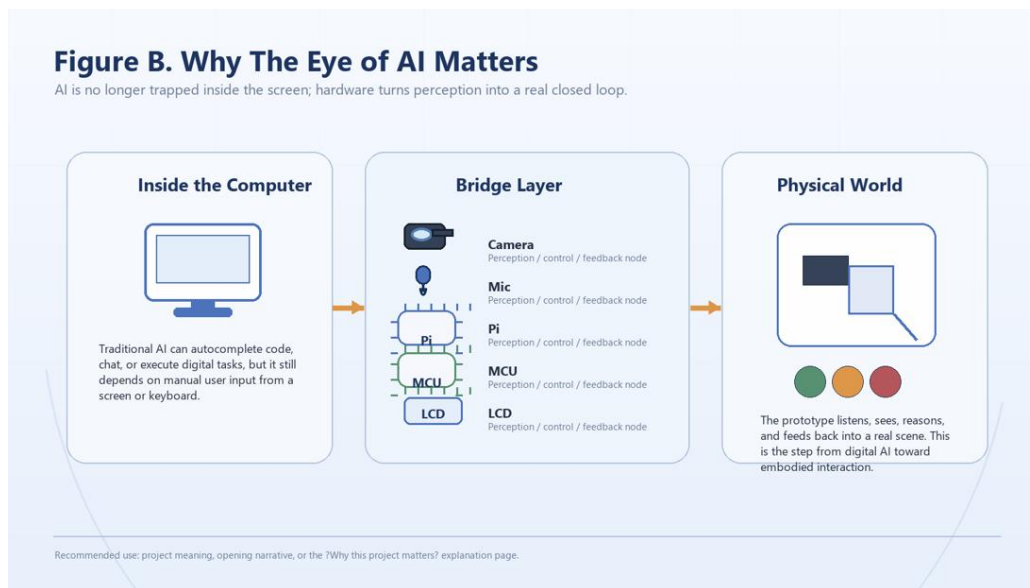
| 栏目                                                            | 建议                       |
|---------------------------------------------------------------|--------------------------|
| figure_A_evolution_timeline.png (封面后 / Slide 4 背景动机)          | 建立 AI 演化主线               |
| figure_B_bridge_significance.png (Slide 1 / Slide 4 / UI 相关页) | 强调 AI 与现实之间的桥梁意义         |
| figure_C_prototype_vs_product.png (Slide 6)                   | 压住 ‘为什么不是成品眼镜’ 的追问       |
| figure_D_closed_loop_architecture.png (Slide 5 / 8 / 10)      | 最强系统主图                   |
| figure_E_workflow_fallback.png (Slide 11 / Appendix)          | 单次演示流程 + fallback        |
| figure_F_boundary_map.png (Slide 15)                          | 把限制讲成 prototype boundary |
| figure_G_future_roadmap.png (Slide 18)                        | 结尾升华与未来工作                |

figure\_A\_evolution\_timeline.png



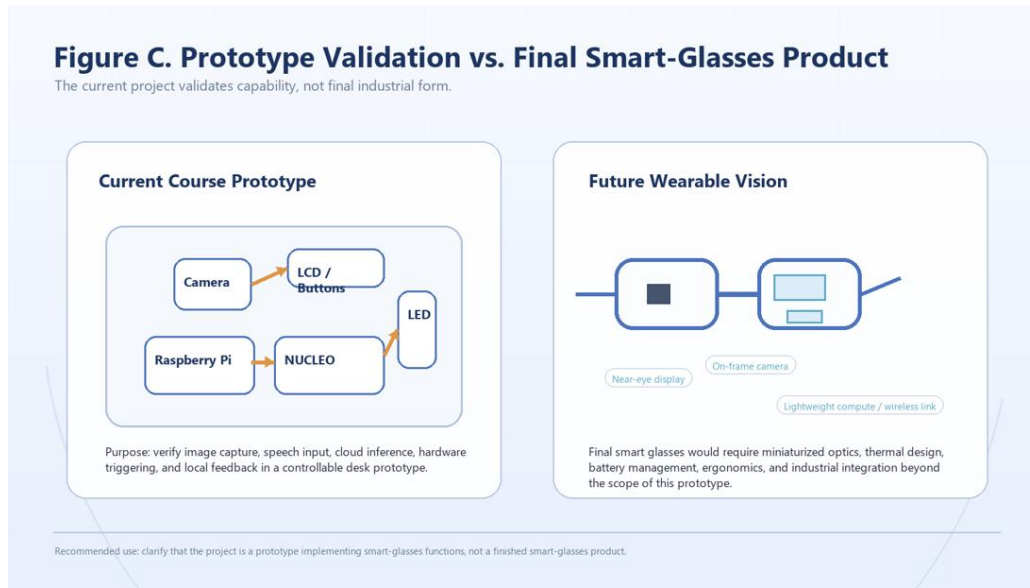
建议插入位置：封面后 / Slide 4 背景动机 | 用途：建立 AI 演化主线

figure\_B\_bridge\_significance.png



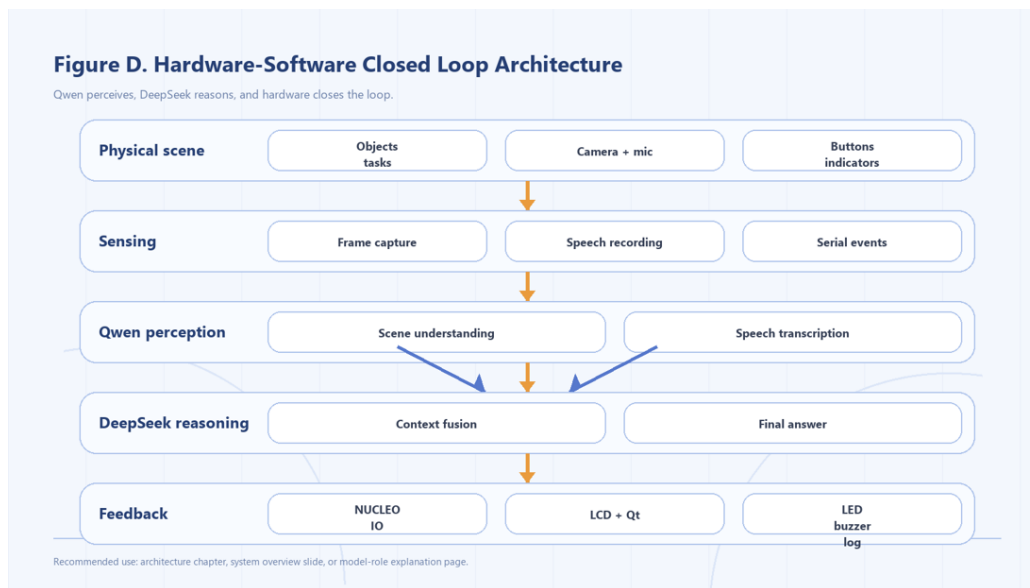
建议插入位置：Slide 1 / Slide 4 / UI 相关页 | 用途：强调 AI 与现实之间的桥梁意义

figure\_C\_prototype\_vs\_product.png



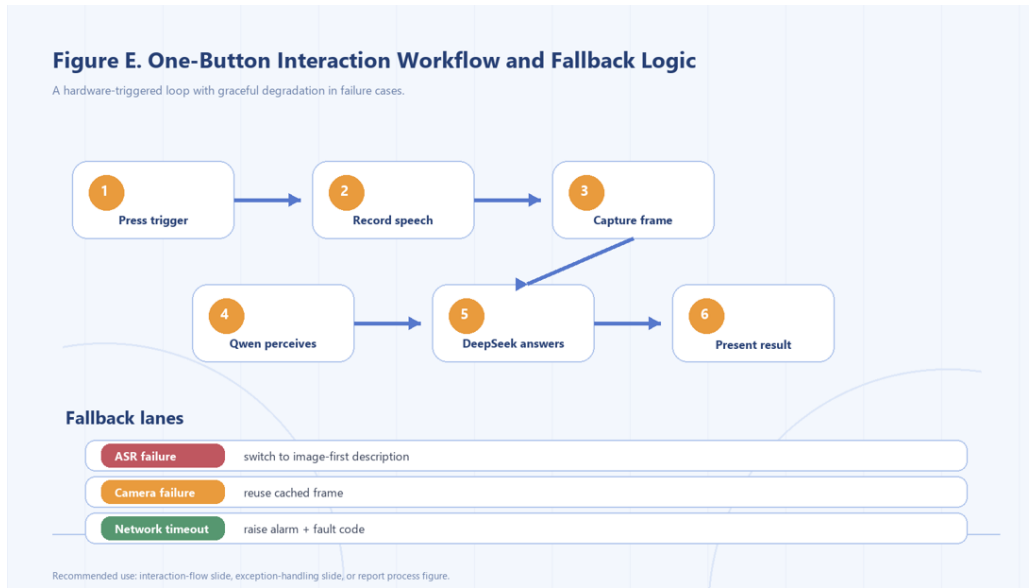
建议插入位置: Slide 6 | 用途: 压住 ‘为什么不是成品眼镜’ 的追问

figure\_D\_closed\_loop\_architecture.png



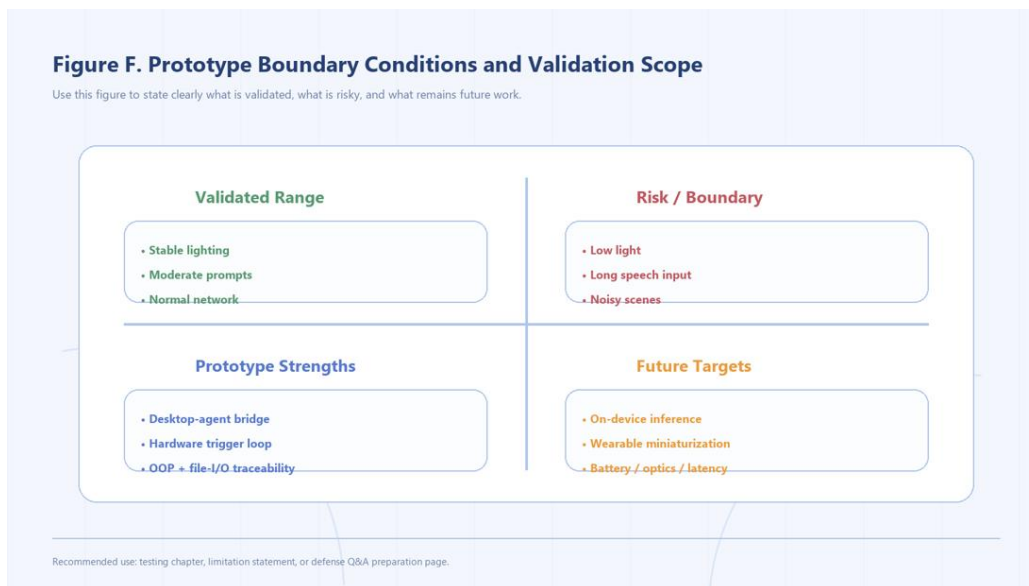
建议插入位置: Slide 5 / 8 / 10 | 用途: 最强系统主图

figure\_E\_workflow\_fallback.png



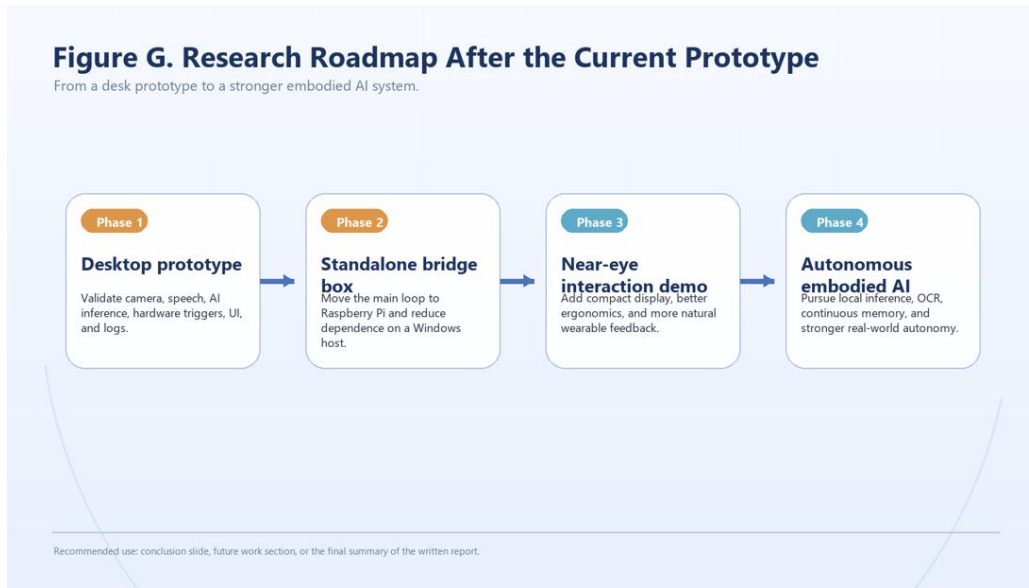
建议插入位置：Slide 11 / Appendix | 用途：单次演示流程 + fallback

figure\_F\_boundary\_map.png



建议插入位置：Slide 15 | 用途：把限制讲成 prototype boundary

figure\_G\_future\_roadmap.png



建议插入位置：Slide 18 | 用途：结尾升华与未来工作